

Create static employee list data using spring xml configuration

1. src/main/java/com/cognizant/spring_learn/model/Skill.java

```
java

package com.cognizant.spring_learn.model;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class Skill {

    private static final Logger LOGGER = LoggerFactory.getLogger(Skill.class);

    private int id;
    private String name;

    public Skill() {
        LOGGER.debug("Inside Skill Constructor.");
    }

    public int getId() {
        LOGGER.debug("Inside getId() method. Value : {}", id);
        return id;
    }

    public void setId(int id) {
        LOGGER.debug("Inside setId() method. Value : {}", id);
        this.id = id;
    }
}
```

```

public String getName() {
    LOGGER.debug("Inside getName() method. Value : {}", name);
    return name;
}

public void setName(String name) {
    LOGGER.debug("Inside setName() method. Value : {}", name);
    this.name = name;
}

@Override
public String toString() {
    return "Skill [id=" + id + ", name=" + name + "]";
}
}

```

2. src/main/java/com/cognizant/spring_learn/model/Department.java

```

java
package com.cognizant.spring_learn.model;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class Department {

    private static final Logger LOGGER = LoggerFactory.getLogger(Department.class);

```

```
private int id;
```

```
private String name;
```

```
public Department() {
```

```
    LOGGER.debug("Inside Department Constructor.");
```

```
}
```

```
public int getId() {
```

```
    LOGGER.debug("Inside getId() method. Value : {}", id);
```

```
    return id;
```

```
}
```

```
public void setId(int id) {
```

```
    LOGGER.debug("Inside setId() method. Value : {}", id);
```

```
    this.id = id;
```

```
}
```

```
public String getName() {
```

```
    LOGGER.debug("Inside getName() method. Value : {}", name);
```

```
    return name;
```

```
}
```

```
public void setName(String name) {
```

```
    LOGGER.debug("Inside setName() method. Value : {}", name);
```

```
    this.name = name;
```

```
}
```

```
@Override
```

```
public String toString() {  
    return "Department [id=" + id + ", name=" + name + "];"  
}  
}
```

3. src/main/java/com/cognizant/spring_learn/model/Employee.java

java

```
package com.cognizant.spring_learn.model;
```

```
import java.util.List;
```

```
import org.slf4j.Logger;
```

```
import org.slf4j.LoggerFactory;
```

```
public class Employee {
```

```
    private static final Logger LOGGER = LoggerFactory.getLogger(Employee.class);
```

```
    private int id;
```

```
    private String firstName;
```

```
    private String lastName;
```

```
    private String email;
```

```
    private String dateOfJoining;
```

```
    private Department department;
```

```
    private List<Skill> skills;
```

```
    public Employee() {
```

```
        LOGGER.debug("Inside Employee Constructor.");
```

```
}
```

```
public int getId() {  
    LOGGER.debug("Inside getId() method. Value : {}", id);  
    return id;  
}
```

```
public void setId(int id) {  
    LOGGER.debug("Inside setId() method. Value : {}", id);  
    this.id = id;  
}
```

```
public String getFirstName() {  
    LOGGER.debug("Inside getFirstName() method. Value : {}", firstName);  
    return firstName;  
}
```

```
public void setFirstName(String firstName) {  
    LOGGER.debug("Inside setFirstName() method. Value : {}", firstName);  
    this.firstName = firstName;  
}
```

```
public String getLastName() {  
    LOGGER.debug("Inside getLastName() method. Value : {}", lastName);  
    return lastName;  
}
```

```
public void setLastName(String lastName) {
```

```
    LOGGER.debug("Inside setLastName() method. Value : {}", lastName);

    this.lastName = lastName;
}

public String getEmail() {

    LOGGER.debug("Inside getEmail() method. Value : {}", email);

    return email;
}

public void setEmail(String email) {

    LOGGER.debug("Inside setEmail() method. Value : {}", email);

    this.email = email;
}

public String getDateOfJoining() {

    LOGGER.debug("Inside getDateOfJoining() method. Value : {}", dateOfJoining);

    return dateOfJoining;
}

public void setDateOfJoining(String dateOfJoining) {

    LOGGER.debug("Inside setDateOfJoining() method. Value : {}", dateOfJoining);

    this.dateOfJoining = dateOfJoining;
}

public Department getDepartment() {

    LOGGER.debug("Inside getDepartment() method. Value : {}", department);

    return department;
}
```

```

public void setDepartment(Department department) {
    LOGGER.debug("Inside setDepartment() method. Value : {}", department);
    this.department = department;
}

```

```

public List<Skill> getSkills() {
    LOGGER.debug("Inside getSkills() method. Value : {}", skills);
    return skills;
}

```

```

public void setSkills(List<Skill> skills) {
    LOGGER.debug("Inside setSkills() method. Value : {}", skills);
    this.skills = skills;
}

```

@Override

```

public String toString() {
    return "Employee [id=" + id + ", firstName=" + firstName + ", lastName=" + lastName
        + ", email=" + email + ", dateOfJoining=" + dateOfJoining
        + ", department=" + department + ", skills=" + skills + "];"
}
}

```

4. src/main/resources/employee.xml

xml

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<beans xmlns="http://www.springframework.org/schema/beans"
```

```
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">
```

```
<!-- Departments -->
```

```
<bean id="itDept" class="com.cognizant.spring_learn.model.Department">
    <property name="id" value="1" />
    <property name="name" value="Information Technology" />
</bean>
```

```
<bean id="hrDept" class="com.cognizant.spring_learn.model.Department">
    <property name="id" value="2" />
    <property name="name" value="Human Resources" />
</bean>
```

```
<bean id="financeDept" class="com.cognizant.spring_learn.model.Department">
    <property name="id" value="3" />
    <property name="name" value="Finance" />
</bean>
```

```
<!-- Skills -->
```

```
<bean id="javaSkill" class="com.cognizant.spring_learn.model.Skill">
    <property name="id" value="1" />
    <property name="name" value="Java" />
</bean>
```

```
<bean id="angularSkill" class="com.cognizant.spring_learn.model.Skill">
    <property name="id" value="2" />
```



```
<property name="name" value="Angular" />
</bean>

<bean id="springSkill" class="com.cognizant.spring_learn.model.Skill">
    <property name="id" value="3" />
    <property name="name" value="Spring Boot" />
</bean>

<bean id="sqlSkill" class="com.cognizant.spring_learn.model.Skill">
    <property name="id" value="4" />
    <property name="name" value="SQL" />
</bean>

<!-- Employee 1 -->
<bean id="employee1" class="com.cognizant.spring_learn.model.Employee">
    <property name="id" value="101" />
    <property name="firstName" value="John" />
    <property name="lastName" value="Doe" />
    <property name="email" value="john.doe@cognizant.com" />
    <property name="dateOfJoining" value="2018-03-15" />
    <property name="department" ref="itDept" />
    <property name="skills">
        <list>
            <ref bean="javaSkill" />
            <ref bean="springSkill" />
        </list>
    </property>
</bean>
```

<!-- Employee 2 -->

<bean id="employee2" class="com.cognizant.spring_learn.model.Employee">

<property name="id" value="102" />

<property name="firstName" value="Jane" />

<property name="lastName" value="Smith" />

<property name="email" value="jane.smith@cognizant.com" />

<property name="dateOfJoining" value="2019-07-22" />

<property name="department" ref="itDept" />

<property name="skills">

<list>

<ref bean="angularSkill" />

<ref bean="javaSkill" />

</list>

</property>

</bean>

<!-- Employee 3 -->

<bean id="employee3" class="com.cognizant.spring_learn.model.Employee">

<property name="id" value="103" />

<property name="firstName" value="Robert" />

<property name="lastName" value="Brown" />

<property name="email" value="robert.brown@cognizant.com" />

<property name="dateOfJoining" value="2017-11-05" />

<property name="department" ref="hrDept" />

<property name="skills">

<list>

<ref bean="sqlSkill" />

```
        </list>
    </property>
</bean>
```

```
<!-- Employee 4 -->
```

```
<bean id="employee4" class="com.cognizant.spring_learn.model.Employee">
    <property name="id" value="104" />
    <property name="firstName" value="Emily" />
    <property name="lastName" value="Davis" />
    <property name="email" value="emily.davis@cognizant.com" />
    <property name="dateOfJoining" value="2020-01-10" />
    <property name="department" ref="financeDept" />
    <property name="skills">
        <list>
            <ref bean="sqlSkill" />
            <ref bean="springSkill" />
        </list>
    </property>
</bean>
```

```
<!-- Employee list as ArrayList (core beans namespace only, no util schema needed) -->
```

```
<bean id="employeeList" class="java.util.ArrayList">
    <constructor-arg>
        <list>
            <ref bean="employee1" />
            <ref bean="employee2" />
            <ref bean="employee3" />
        </list>
    </constructor-arg>
</bean>
```

```
        <ref bean="employee4" />

    </list>

</constructor-arg>

</bean>

</beans>
```

5. src/main/java/com/cognizant/spring_learn/dao/EmployeeDao.java

```
java

package com.cognizant.spring_learn.dao;

import java.util.ArrayList;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.stereotype.Repository;

import com.cognizant.spring_learn.model.Employee;

@Repository

public class EmployeeDao {

    private static final Logger LOGGER = LoggerFactory.getLogger(EmployeeDao.class);

    private static ArrayList<Employee> EMPLOYEE_LIST;
```

```

@SuppressWarnings("unchecked")

public EmployeeDao() {

    LOGGER.debug("Inside EmployeeDao Constructor.");

    ApplicationContext context = new
    ClassPathXmlApplicationContext("employee.xml");

    EMPLOYEE_LIST = (ArrayList<Employee>) context.getBean("employeeList",
    ArrayList.class);

    LOGGER.debug("EMPLOYEE_LIST loaded with {} employees.",
    EMPLOYEE_LIST.size());

}

public ArrayList<Employee> getAllEmployees() {

    LOGGER.debug("Start :: getAllEmployees()");

    LOGGER.debug("End :: getAllEmployees() returning {} employees.",
    EMPLOYEE_LIST.size());

    return EMPLOYEE_LIST;

}

}

```

```

2026-07-08 13:21:34.656 INFO 2644 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port(s): 8090 (http)
2026-07-08 13:21:34.663 INFO 2644 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-08 13:21:34.663 INFO 2644 --- [main] org.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/9.0.83]
2026-07-08 13:21:34.747 INFO 2644 --- [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-07-08 13:21:34.747 INFO 2644 --- [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 4
2026-07-08 13:21:34.791 DEBUG 2644 --- [main] c.c.spring_learn.dao.EmployeeDao : Inside EmployeeDao Constructor.
2026-07-08 13:21:34.883 DEBUG 2644 --- [main] c.c.spring_learn.model.Department : Inside Department Constructor.
2026-07-08 13:21:34.884 DEBUG 2644 --- [main] c.c.spring_learn.model.Department : Inside setId() method. Value : 1
2026-07-08 13:21:34.885 DEBUG 2644 --- [main] c.c.spring_learn.model.Department : Inside setName() method. Value : Information Technology
2026-07-08 13:21:34.885 DEBUG 2644 --- [main] c.c.spring_learn.model.Department : Inside Department Constructor.
2026-07-08 13:21:34.885 DEBUG 2644 --- [main] c.c.spring_learn.model.Department : Inside setId() method. Value : 2
2026-07-08 13:21:34.885 DEBUG 2644 --- [main] c.c.spring_learn.model.Department : Inside setName() method. Value : Human Resources
2026-07-08 13:21:34.885 DEBUG 2644 --- [main] c.c.spring_learn.model.Department : Inside Department Constructor.
2026-07-08 13:21:34.886 DEBUG 2644 --- [main] c.c.spring_learn.model.Department : Inside setId() method. Value : 3
2026-07-08 13:21:34.886 DEBUG 2644 --- [main] c.c.spring_learn.model.Department : Inside setName() method. Value : Finance
2026-07-08 13:21:34.886 DEBUG 2644 --- [main] com.cognizant.spring_learn.model.Skill : Inside Skill Constructor.
2026-07-08 13:21:34.887 DEBUG 2644 --- [main] com.cognizant.spring_learn.model.Skill : Inside setId() method. Value : 1
2026-07-08 13:21:34.887 DEBUG 2644 --- [main] com.cognizant.spring_learn.model.Skill : Inside setName() method. Value : Java
2026-07-08 13:21:34.887 DEBUG 2644 --- [main] com.cognizant.spring_learn.model.Skill : Inside Skill Constructor.
2026-07-08 13:21:34.887 DEBUG 2644 --- [main] com.cognizant.spring_learn.model.Skill : Inside setId() method. Value : 2
2026-07-08 13:21:34.887 DEBUG 2644 --- [main] com.cognizant.spring_learn.model.Skill : Inside setName() method. Value : Angular
2026-07-08 13:21:34.887 DEBUG 2644 --- [main] com.cognizant.spring_learn.model.Skill : Inside Skill Constructor.
2026-07-08 13:21:34.887 DEBUG 2644 --- [main] com.cognizant.spring_learn.model.Skill : Inside setId() method. Value : 3
2026-07-08 13:21:34.887 DEBUG 2644 --- [main] com.cognizant.spring_learn.model.Skill : Inside setName() method. Value : Spring Boot
2026-07-08 13:21:34.887 DEBUG 2644 --- [main] com.cognizant.spring_learn.model.Skill : Inside Skill Constructor.
2026-07-08 13:21:34.887 DEBUG 2644 --- [main] com.cognizant.spring_learn.model.Skill : Inside setId() method. Value : 4

```



```
<version>1.0-SNAPSHOT</version>
```

```
<packaging>jar</packaging>
```

```
<properties>
```

```
  <java.version>1.8</java.version>
```

```
</properties>
```

```
<dependencies>
```

```
  <!-- Spring Boot Web (includes Spring MVC + embedded Tomcat) -->
```

```
  <dependency>
```

```
    <groupId>org.springframework.boot</groupId>
```

```
    <artifactId>spring-boot-starter-web</artifactId>
```

```
  </dependency>
```

```
  <!-- Log4j2 logging -->
```

```
  <dependency>
```

```
    <groupId>org.springframework.boot</groupId>
```

```
    <artifactId>spring-boot-starter-log4j2</artifactId>
```

```
  </dependency>
```

```
  <!-- Spring Security -->
```

```
  <dependency>
```

```
    <groupId>org.springframework.boot</groupId>
```

```
    <artifactId>spring-boot-starter-security</artifactId>
```

```
  </dependency>
```

```
  <!-- Spring Transaction support (for @Transactional) -->
```

```
  <dependency>
```

```
<groupId>org.springframework</groupId>
<artifactId>spring-tx</artifactId>
</dependency>

<!-- JWT (jjwt) -->
<dependency>
  <groupId>io.jsonwebtoken</groupId>
  <artifactId>jjwt-api</artifactId>
  <version>0.11.5</version>
</dependency>
<dependency>
  <groupId>io.jsonwebtoken</groupId>
  <artifactId>jjwt-impl</artifactId>
  <version>0.11.5</version>
  <scope>runtime</scope>
</dependency>
<dependency>
  <groupId>io.jsonwebtoken</groupId>
  <artifactId>jjwt-jackson</artifactId>
  <version>0.11.5</version>
  <scope>runtime</scope>
</dependency>
</dependencies>

<build>
  <plugins>
    <plugin>
      <groupId>org.springframework.boot</groupId>
```



```
        <artifactId>spring-boot-maven-plugin</artifactId>
    </plugin>
</plugins>
</build>
</project>
```

for all

Identified request to reproduce three code files

Identified request to reproduce three code files

File: pom.xml

xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>

    <parent>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-parent</artifactId>
        <version>2.7.18</version>
        <relativePath/>
    </parent>

    <groupId>com.cognizant.springlearn</groupId>
    <artifactId>SpringLearnApp</artifactId>
    <version>1.0-SNAPSHOT</version>
    <packaging>jar</packaging>
```

<properties>

 <java.version>1.8</java.version>

</properties>

<dependencies>

 <!-- Spring Boot Web (includes Spring MVC + embedded Tomcat) -->

 <dependency>

 <groupId>org.springframework.boot</groupId>

 <artifactId>spring-boot-starter-web</artifactId>

 </dependency>

 <!-- Log4j2 logging -->

 <dependency>

 <groupId>org.springframework.boot</groupId>

 <artifactId>spring-boot-starter-log4j2</artifactId>

 </dependency>

 <!-- Spring Security -->

 <dependency>

 <groupId>org.springframework.boot</groupId>

 <artifactId>spring-boot-starter-security</artifactId>

 </dependency>

 <!-- Spring Transaction support (for @Transactional) -->

 <dependency>

 <groupId>org.springframework</groupId>

 <artifactId>spring-tx</artifactId>

</dependency>

<!-- JWT (jjwt) -->

<dependency>

<groupId>io.jsonwebtoken</groupId>

<artifactId>jjwt-api</artifactId>

<version>0.11.5</version>

</dependency>

<dependency>

<groupId>io.jsonwebtoken</groupId>

<artifactId>jjwt-impl</artifactId>

<version>0.11.5</version>

<scope>runtime</scope>

</dependency>

<dependency>

<groupId>io.jsonwebtoken</groupId>

<artifactId>jjwt-jackson</artifactId>

<version>0.11.5</version>

<scope>runtime</scope>

</dependency>

</dependencies>

<build>

<plugins>

<plugin>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-maven-plugin</artifactId>

</plugin>

```
</plugins>

</build>

</project>
```

File: src/main/java/com/cognizant/spring_learn/service/EmployeeService.java

```
java
```

```
package com.cognizant.spring_learn.service;
```

```
import java.util.ArrayList;
```

```
import org.slf4j.Logger;
```

```
import org.slf4j.LoggerFactory;
```

```
import org.springframework.beans.factory.annotation.Autowired;
```

```
import org.springframework.stereotype.Service;
```

```
import org.springframework.transaction.annotation.Transactional;
```

```
import com.cognizant.spring_learn.dao.EmployeeDao;
```

```
import com.cognizant.spring_learn.model.Employee;
```

```
@Service
```

```
public class EmployeeService {
```

```
    private static final Logger LOGGER =
    LoggerFactory.getLogger(EmployeeService.class);
```

```
    @Autowired
```

```
    private EmployeeDao employeeDao;
```

```

@Transactional

public ArrayList<Employee> getAllEmployees() {

    LOGGER.debug("Start :: getAllEmployees()");

    ArrayList<Employee> employeeList = employeeDao.getAllEmployees();

    LOGGER.debug("End :: getAllEmployees() returning {} employees.",
employeeList.size());

    return employeeList;

}
}

```

File:

src/main/java/com/cognizant/spring_learn/controller/EmployeeController.java

java

package com.cognizant.spring_learn.controller;

import java.util.ArrayList;

import org.slf4j.Logger;

import org.slf4j.LoggerFactory;

import org.springframework.beans.factory.annotation.Autowired;

import org.springframework.web.bind.annotation.GetMapping;

import org.springframework.web.bind.annotation.RestController;

import com.cognizant.spring_learn.model.Employee;

import com.cognizant.spring_learn.service.EmployeeService;

@RestController

```
public class EmployeeController {

    private static final Logger LOGGER =
LoggerFactory.getLogger(EmployeeController.class);

    @Autowired

    private EmployeeService employeeService;

    @GetMapping("/employees")
    public ArrayList<Employee> getAllEmployees() {
        LOGGER.debug("Start :: getAllEmployees()");

        ArrayList<Employee> employeeList = employeeService.getAllEmployees();

        LOGGER.debug("End :: getAllEmployees() returning {} employees.",
employeeList.size());

        return employeeList;
    }
}
```



```
2026-07-08 13:27:56.465 DEBUG 11616 --- [main] com.cognizant.spring_learn.model.Skill : Inside setId() method. Value : 4
2026-07-08 13:27:56.465 DEBUG 11616 --- [main] com.cognizant.spring_learn.model.Skill : Inside setName() method. Value : SQL
2026-07-08 13:27:56.465 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside Employee Constructor.
2026-07-08 13:27:56.467 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setId() method. Value : 101
2026-07-08 13:27:56.467 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setFirstName() method. Value : John
2026-07-08 13:27:56.467 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setLastName() method. Value : Doe
2026-07-08 13:27:56.467 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setEmail() method. Value : john.doe@cognizant.com
2026-07-08 13:27:56.467 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setDateOfJoining() method. Value : 2018-03-15
2026-07-08 13:27:56.468 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setDepartment() method. Value : Department [id=1, name=Information
2026-07-08 13:27:56.469 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setSkills() method. Value : [Skill [id=1, name=Java], Skill [id=3,
2026-07-08 13:27:56.469 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside Employee Constructor.
2026-07-08 13:27:56.469 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setId() method. Value : 102
2026-07-08 13:27:56.469 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setFirstName() method. Value : Jane
2026-07-08 13:27:56.469 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setLastName() method. Value : Smith
2026-07-08 13:27:56.469 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setEmail() method. Value : jane.smith@cognizant.com
2026-07-08 13:27:56.470 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setDateOfJoining() method. Value : 2019-07-22
2026-07-08 13:27:56.470 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setDepartment() method. Value : Department [id=1, name=Information
2026-07-08 13:27:56.470 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setSkills() method. Value : [Skill [id=2, name=Angular], Skill [id=
2026-07-08 13:27:56.470 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside Employee Constructor.
2026-07-08 13:27:56.470 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setId() method. Value : 103
2026-07-08 13:27:56.470 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setFirstName() method. Value : Robert
2026-07-08 13:27:56.470 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setLastName() method. Value : Brown
2026-07-08 13:27:56.470 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setEmail() method. Value : robert.brown@cognizant.com
2026-07-08 13:27:56.470 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setDateOfJoining() method. Value : 2017-11-05
2026-07-08 13:27:56.470 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setDepartment() method. Value : Department [id=2, name=Human Resourc
2026-07-08 13:27:56.470 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setSkills() method. Value : [Skill [id=4, name=SQL]]
2026-07-08 13:27:56.470 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside Employee Constructor.
2026-07-08 13:27:56.470 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setId() method. Value : 104
```

```
2026-07-08 13:27:56.472 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setEmail() method. Value : emily.davis@cognizant.com
2026-07-08 13:27:56.472 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setDateOfJoining() method. Value : 2020-01-10
2026-07-08 13:27:56.472 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setDepartment() method. Value : Department [id=3, name=Finance]
2026-07-08 13:27:56.472 DEBUG 11616 --- [main] c.cognizant.spring_learn.model.Employee : Inside setSkills() method. Value : [Skill [id=4, name=SQL], Skill [id=3, n
2026-07-08 13:27:56.479 DEBUG 11616 --- [main] c.c.spring_learn.dao.EmployeeDao : EMPLOYEE_LIST loaded with 4 employees.
2026-07-08 13:27:56.912 INFO 11616 --- [main] o.s.s.web.DefaultSecurityFilterChain : Will secure any request with [org.springframework.security.web.session.Dis
2026-07-08 13:27:56.955 INFO 11616 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port(s): 8090 (http) with context path '/'
2026-07-08 13:27:56.960 INFO 11616 --- [main] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 2.351 seconds (JVM running for 2.819)
2026-07-08 13:28:01.914 INFO 11616 --- [nio-8090-exec-1] o.a.c.e.g.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-08 13:28:01.915 INFO 11616 --- [nio-8090-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-08 13:28:01.915 INFO 11616 --- [nio-8090-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 0 ms
2026-07-08 13:28:02.272 DEBUG 11616 --- [nio-8090-exec-1] c.c.s.controller.EmployeeController : Start :: getAllEmployees()
2026-07-08 13:28:02.272 DEBUG 11616 --- [nio-8090-exec-1] c.c.s.service.EmployeeService : Start :: getAllEmployees()
2026-07-08 13:28:02.292 DEBUG 11616 --- [nio-8090-exec-1] c.c.spring_learn.dao.EmployeeDao : Start :: getAllEmployees()
2026-07-08 13:28:02.293 DEBUG 11616 --- [nio-8090-exec-1] c.c.spring_learn.dao.EmployeeDao : End :: getAllEmployees() returning 4 employees.
2026-07-08 13:28:02.293 DEBUG 11616 --- [nio-8090-exec-1] c.c.s.service.EmployeeService : End :: getAllEmployees() returning 4 employees.
2026-07-08 13:28:02.293 DEBUG 11616 --- [nio-8090-exec-1] c.c.s.controller.EmployeeController : End :: getAllEmployees() returning 4 employees.
2026-07-08 13:28:02.372 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getId() method. Value : 101
2026-07-08 13:28:02.373 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getFirstName() method. Value : John
2026-07-08 13:28:02.373 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getLastName() method. Value : Doe
2026-07-08 13:28:02.373 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getEmail() method. Value : john.doe@cognizant.com
2026-07-08 13:28:02.374 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getDateOfJoining() method. Value : 2018-03-15
2026-07-08 13:28:02.374 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getDepartment() method. Value : Department [id=1, name=Information
2026-07-08 13:28:02.374 DEBUG 11616 --- [nio-8090-exec-1] c.c.spring_learn.model.Department : Inside getId() method. Value : 1
2026-07-08 13:28:02.374 DEBUG 11616 --- [nio-8090-exec-1] c.c.spring_learn.model.Department : Inside getName() method. Value : Information Technology
2026-07-08 13:28:02.375 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getSkills() method. Value : [Skill [id=1, name=Java], Skill [id=3,
2026-07-08 13:28:02.375 DEBUG 11616 --- [nio-8090-exec-1] com.cognizant.spring_learn.model.Skill : Inside getId() method. Value : 1
2026-07-08 13:28:02.376 DEBUG 11616 --- [nio-8090-exec-1] com.cognizant.spring_learn.model.Skill : Inside getName() method. Value : Java
2026-07-08 13:28:02.376 DEBUG 11616 --- [nio-8090-exec-1] com.cognizant.spring_learn.model.Skill : Inside getId() method. Value : 3
```

```
2026-07-08 13:28:02.376 DEBUG 11616 --- [nio-8090-exec-1] com.cognizant.spring_learn.model.Skill : Inside getName() method. Value : Angular
2026-07-08 13:28:02.377 DEBUG 11616 --- [nio-8090-exec-1] com.cognizant.spring_learn.model.Skill : Inside getId() method. Value : 1
2026-07-08 13:28:02.377 DEBUG 11616 --- [nio-8090-exec-1] com.cognizant.spring_learn.model.Skill : Inside getName() method. Value : Java
2026-07-08 13:28:02.377 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getId() method. Value : 103
2026-07-08 13:28:02.377 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getFirstName() method. Value : Robert
2026-07-08 13:28:02.377 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getLastName() method. Value : Brown
2026-07-08 13:28:02.377 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getEmail() method. Value : robert.brown@cognizant.com
2026-07-08 13:28:02.377 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getDateOfJoining() method. Value : 2017-11-05
2026-07-08 13:28:02.377 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getDepartment() method. Value : Department [id=2, name=Human Reso
2026-07-08 13:28:02.377 DEBUG 11616 --- [nio-8090-exec-1] c.c.spring_learn.model.Department : Inside getId() method. Value : 2
2026-07-08 13:28:02.377 DEBUG 11616 --- [nio-8090-exec-1] c.c.spring_learn.model.Department : Inside getName() method. Value : Human Resources
2026-07-08 13:28:02.377 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getSkills() method. Value : [Skill [id=4, name=SQL]]
2026-07-08 13:28:02.377 DEBUG 11616 --- [nio-8090-exec-1] com.cognizant.spring_learn.model.Skill : Inside getId() method. Value : 4
2026-07-08 13:28:02.378 DEBUG 11616 --- [nio-8090-exec-1] com.cognizant.spring_learn.model.Skill : Inside getName() method. Value : SQL
2026-07-08 13:28:02.378 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getId() method. Value : 104
2026-07-08 13:28:02.378 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getFirstName() method. Value : Emily
2026-07-08 13:28:02.378 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getLastName() method. Value : Davis
2026-07-08 13:28:02.378 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getEmail() method. Value : emily.davis@cognizant.com
2026-07-08 13:28:02.378 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getDateOfJoining() method. Value : 2020-01-10
2026-07-08 13:28:02.378 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getDepartment() method. Value : Department [id=3, name=Finance]
2026-07-08 13:28:02.378 DEBUG 11616 --- [nio-8090-exec-1] c.c.spring_learn.model.Department : Inside getId() method. Value : 3
2026-07-08 13:28:02.378 DEBUG 11616 --- [nio-8090-exec-1] c.c.spring_learn.model.Department : Inside getName() method. Value : Finance
2026-07-08 13:28:02.378 DEBUG 11616 --- [nio-8090-exec-1] c.cognizant.spring_learn.model.Employee : Inside getSkills() method. Value : [Skill [id=4, name=SQL], Skill [id=3,
2026-07-08 13:28:02.378 DEBUG 11616 --- [nio-8090-exec-1] com.cognizant.spring_learn.model.Skill : Inside getId() method. Value : 4
2026-07-08 13:28:02.378 DEBUG 11616 --- [nio-8090-exec-1] com.cognizant.spring_learn.model.Skill : Inside getName() method. Value : SQL
2026-07-08 13:28:02.378 DEBUG 11616 --- [nio-8090-exec-1] com.cognizant.spring_learn.model.Skill : Inside getId() method. Value : 3
2026-07-08 13:28:02.378 DEBUG 11616 --- [nio-8090-exec-1] com.cognizant.spring_learn.model.Skill : Inside getName() method. Value : Spring Boot
```


Create REST service for department

File: src/main/resources/department.xml

xml

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<beans xmlns="http://www.springframework.org/schema/beans"
```

```
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
    xsi:schemaLocation="http://www.springframework.org/schema/beans
```

```
    http://www.springframework.org/schema/beans/spring-beans.xsd">
```

```
<bean id="itDept" class="com.cognizant.spring_learn.model.Department">
```

```
    <property name="id" value="1" />
```

```
    <property name="name" value="Information Technology" />
```

```
</bean>
```

```
<bean id="hrDept" class="com.cognizant.spring_learn.model.Department">
```

```
    <property name="id" value="2" />
```

```
    <property name="name" value="Human Resources" />
```

```
</bean>
```

```
<bean id="financeDept" class="com.cognizant.spring_learn.model.Department">
```

```
    <property name="id" value="3" />
```

```
    <property name="name" value="Finance" />
```

```
</bean>
```

```
<bean id="departmentList" class="java.util.ArrayList">
```

```
    <constructor-arg>
```

```
        <list>
```

```
            <ref bean="itDept" />
```

```
        <ref bean="hrDept" />

        <ref bean="financeDept" />

    </list>

</constructor-arg>

</bean>

</beans>
```

File: src/main/java/com/cognizant/spring_learn/dao/DepartmentDao.java

```
java
```

```
package com.cognizant.spring_learn.dao;
```

```
import java.util.ArrayList;
```

```
import org.slf4j.Logger;
```

```
import org.slf4j.LoggerFactory;
```

```
import org.springframework.context.ApplicationContext;
```

```
import org.springframework.context.support.ClassPathXmlApplicationContext;
```

```
import org.springframework.stereotype.Repository;
```

```
import com.cognizant.spring_learn.model.Department;
```

```
@Repository
```

```
public class DepartmentDao {
```

```
    private static final Logger LOGGER = LoggerFactory.getLogger(DepartmentDao.class);
```

```
    private static ArrayList<Department> DEPARTMENT_LIST;
```

```

@SuppressWarnings("unchecked")

public DepartmentDao() {

    LOGGER.debug("Inside DepartmentDao Constructor.");

    ApplicationContext context = new
    ClassPathXmlApplicationContext("department.xml");

    DEPARTMENT_LIST = (ArrayList<Department>) context.getBean("departmentList",
    ArrayList.class);

    LOGGER.debug("DEPARTMENT_LIST loaded with {} departments.",
    DEPARTMENT_LIST.size());

}

public ArrayList<Department> getAllDepartments() {

    LOGGER.debug("Start :: getAllDepartments()");

    LOGGER.debug("End :: getAllDepartments() returning {} departments.",
    DEPARTMENT_LIST.size());

    return DEPARTMENT_LIST;

}

}

```

File: src/main/java/com/cognizant/spring_learn/service/DepartmentService.java

```

java

package com.cognizant.spring_learn.service;

import java.util.ArrayList;

import org.slf4j.Logger;

```

```
import org.slf4j.LoggerFactory;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;


import com.cognizant.spring_learn.dao.DepartmentDao;
import com.cognizant.spring_learn.model.Department;


@Service

public class DepartmentService {


    private static final Logger LOGGER =
LoggerFactory.getLogger(DepartmentService.class);


    @Autowired

    private DepartmentDao departmentDao;


    @Transactional

    public ArrayList<Department> getAllDepartments() {

        LOGGER.debug("Start :: getAllDepartments()");


        ArrayList<Department> departmentList = departmentDao.getAllDepartments();


        LOGGER.debug("End :: getAllDepartments() returning {} departments.",
departmentList.size());

        return departmentList;

    }

}
```

File:**src/main/java/com/cognizant/spring_learn/controller/DepartmentController.java**

```
java
```

```
package com.cognizant.spring_learn.controller;
```

```
import java.util.ArrayList;
```

```
import org.slf4j.Logger;
```

```
import org.slf4j.LoggerFactory;
```

```
import org.springframework.beans.factory.annotation.Autowired;
```

```
import org.springframework.web.bind.annotation.GetMapping;
```

```
import org.springframework.web.bind.annotation.RestController;
```

```
import com.cognizant.spring_learn.model.Department;
```

```
import com.cognizant.spring_learn.service.DepartmentService;
```

```
@RestController
```

```
public class DepartmentController {
```

```
    private static final Logger LOGGER =  
    LoggerFactory.getLogger(DepartmentController.class);
```

```
    @Autowired
```

```
    private DepartmentService departmentService;
```

```
    @GetMapping("/departments")
```

```
    public ArrayList<Department> getAllDepartments() {
```

```
        LOGGER.debug("Start :: getAllDepartments()");
```

```
ArrayList<Department> departmentList = departmentService.getAllDepartments();
```

```
    LOGGER.debug("End :: getAllDepartments() returning {} departments.",  
departmentList.size());
```

```
    return departmentList;
```

```
}
```

```
}
```

HTTP **http://localhost:8090/departments**

GET **http://localhost:8090/departments**

Docs Params **Authorization** Headers 8 Body Scripts Tests Set

Body Cookies 1 Headers 12 Test Results

{ } JSON Preview Visualize

```
1  [
2      {
3          "id": 1,
4          "name": "Information Technology"
5      },
6      {
7          "id": 2,
8          "name": "Human Resources"
9      },
10     {
11         "id": 3,
12         "name": "Finance"
13     }
14 ]
```

```
2026-07-08 13:33:31.929 DEBUG 17252 --- [main] c.cognizant.spring_learn.model.Employee : Inside Employee Constructor.
2026-07-08 13:33:31.929 DEBUG 17252 --- [main] c.cognizant.spring_learn.model.Employee : Inside setId() method. Value : 104
2026-07-08 13:33:31.929 DEBUG 17252 --- [main] c.cognizant.spring_learn.model.Employee : Inside setFirstName() method. Value : Emily
2026-07-08 13:33:31.929 DEBUG 17252 --- [main] c.cognizant.spring_learn.model.Employee : Inside setLastName() method. Value : Davis
2026-07-08 13:33:31.929 DEBUG 17252 --- [main] c.cognizant.spring_learn.model.Employee : Inside setEmail() method. Value : emily.davis@cognizant.com
2026-07-08 13:33:31.929 DEBUG 17252 --- [main] c.cognizant.spring_learn.model.Employee : Inside setDateOfJoining() method. Value : 2020-01-10
2026-07-08 13:33:31.929 DEBUG 17252 --- [main] c.cognizant.spring_learn.model.Employee : Inside setDepartment() method. Value : Department [id=3, name=Finance]
2026-07-08 13:33:31.930 DEBUG 17252 --- [main] c.cognizant.spring_learn.model.Employee : Inside setSkills() method. Value : [Skill [id=4, name=SQL], Skill [id=3, name=Java]]
2026-07-08 13:33:31.931 DEBUG 17252 --- [main] c.c.spring_learn.dao.EmployeeDao : EMPLOYEE_LIST loaded with 4 employees.
2026-07-08 13:33:32.269 INFO 17252 --- [main] o.s.s.web.DefaultSecurityFilterChain : Will secure any request with [org.springframework.security.web.session.DisableSessionCreationServletFilter]
2026-07-08 13:33:32.307 INFO 17252 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port(s): 8090 (http) with context path ''
2026-07-08 13:33:32.311 INFO 17252 --- [main] c.e.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 2.009 seconds (JVM running for 2.468)
2026-07-08 13:33:42.345 INFO 17252 --- [nio-8090-exec-2] o.a.c.c.f.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-08 13:33:42.345 INFO 17252 --- [nio-8090-exec-2] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-08 13:33:42.345 INFO 17252 --- [nio-8090-exec-2] o.s.web.servlet.DispatcherServlet : Completed initialization in 0 ms
2026-07-08 13:33:42.698 DEBUG 17252 --- [nio-8090-exec-2] c.c.s.controller.DepartmentController : Start :: getAllDepartments()
2026-07-08 13:33:42.702 DEBUG 17252 --- [nio-8090-exec-2] c.c.s.service.DepartmentService : Start :: getAllDepartments()
2026-07-08 13:33:42.719 DEBUG 17252 --- [nio-8090-exec-2] c.c.spring_learn.dao.DepartmentDao : Start :: getAllDepartments()
2026-07-08 13:33:42.719 DEBUG 17252 --- [nio-8090-exec-2] c.c.spring_learn.dao.DepartmentDao : End :: getAllDepartments() returning 3 departments.
2026-07-08 13:33:42.719 DEBUG 17252 --- [nio-8090-exec-2] c.c.s.service.DepartmentService : End :: getAllDepartments() returning 3 departments.
2026-07-08 13:33:42.719 DEBUG 17252 --- [nio-8090-exec-2] c.c.s.controller.DepartmentController : End :: getAllDepartments() returning 3 departments.
2026-07-08 13:33:42.783 DEBUG 17252 --- [nio-8090-exec-2] c.c.spring_learn.model.Department : Inside getId() method. Value : 1
2026-07-08 13:33:42.786 DEBUG 17252 --- [nio-8090-exec-2] c.c.spring_learn.model.Department : Inside getName() method. Value : Information Technology
2026-07-08 13:33:42.786 DEBUG 17252 --- [nio-8090-exec-2] c.c.spring_learn.model.Department : Inside getId() method. Value : 2
2026-07-08 13:33:42.786 DEBUG 17252 --- [nio-8090-exec-2] c.c.spring_learn.model.Department : Inside getName() method. Value : Human Resources
2026-07-08 13:33:42.786 DEBUG 17252 --- [nio-8090-exec-2] c.c.spring_learn.model.Department : Inside getId() method. Value : 3
2026-07-08 13:33:42.786 DEBUG 17252 --- [nio-8090-exec-2] c.c.spring_learn.model.Department : Inside getName() method. Value : Finance
```